

Daily Quiz 12 (ELL782/ELL7282) - Computer Architectures

Instructions

- **Time:** 5 minutes
- Maximum marks: 10
- Number of questions: 10
- Each question carries **1 mark**.
- Questions marked **MCQ** have exactly one correct option.
- Questions marked **MSQ** may have one or more correct options. For an MSQ, select all correct options and no incorrect option.
- There is **no partial credit** for an MSQ.

Points: 10/10

Name * 

test

Entry Number * 

test

✓ **Correct** 1/1 Points

Four independent instructions I1, I2, I3, and I4 enter a 5-stage pipeline (IF, OF, EX, MA, RW), one instruction per cycle, with I1 entering IF in cycle 1. What is the stage occupancy in cycle 5? 

- ☒ IF: empty, OF: I4, EX: I3, MA: I2, RW: I1
- ☐ IF: I4, OF: I3, EX: I2, MA: I1, RW: empty

☐ IF: empty, OF: I3, EX: I4, MA: I2, RW: I1

☐ IF: I4, OF: empty, EX: I3, MA: I2, RW: I1

✓ **Correct** 1/1 Points

Which instruction pairs contain a RAW dependency that can create a data hazard in this in-order pipeline?



Select all that apply.

☒ [1] add r1, r2, r3; [2] add r4, r5, r1

☐ [1] add r1, r2, r3; [2] add r2, r5, r6

☒ [1] add r1, r2, r3; [2] sub r4, r1, r5

☐ [1] add r1, r2, r3; [2] sub r1, r4, r5

✓ **Correct** 1/1 Points

Consider the sequence:

[1] add r1, r2, r3

[2] sub r4, r1, r5

With no forwarding and no interlocks, what is the earliest cycle in which instruction [2] can be dispatched safely under the pipeline timing convention used here? 

☒ Cycle 5

☐ Cycle 3

☐ Cycle 2

☐ Cycle 4

✓ **Correct** 1/1 Points

A conditional branch reaches the EX stage and is found to be taken. Which action satisfies the branch-lock condition for the two younger instructions currently in OF and IF? 

- ☐ Allow both instructions to continue, but suppress only their register writes.
- ☐ Convert only the IF-stage instruction to a bubble and let the OF-stage instruction execute.
- ☒ Convert both younger instructions to bubbles and begin fetching from the branch target.
- ☐ Stall the entire pipeline until the branch reaches RW.

✓ **Correct** 1/1 Points

Assume a hypothetical memory-operand instruction: `add r1, r2, 10[r3]` and a single MA resource. Which simultaneous demand is an example of a structural hazard? 

- ☒ A store needs the MA resource for a memory write while the memory-operand add needs the same MA resource for a memory read in the same cycle.
- ☐ An ALU instruction writes a register while a later instruction is still in IF.
- ☐ A branch comparing flags while another instruction is in IF.
- ☐ Two instructions use different registers in the register file in consecutive cycles.

✓ **Correct** 1/1 Points

A delayed branch has exactly two delay slots. The compiler finds only one valid instruction that can safely be moved from before the branch. What should it place in the two delay slots? 

- ☐ Two copies of the valid instruction.
- ☐ The branch target instruction followed by the valid instruction.

- ☒ The valid instruction in one slot and a nop in the other.
- ☐ Two nops, because a partially filled delay region is not allowed.

✓ **Correct** 1/1 Points

A consumer instruction in OF has a RAW conflict with an older instruction in EX/MA/RW. What should a data-lock interlock do?



- ☐ Stall only RW and allow the consumer to enter EX.
- ☐ Stall all five stages and never insert a bubble.
- ☒ Stall IF and OF, and send a bubble (nop packet) into EX.
- ☐ Flush the producer and restart it from IF.

✓ **Correct** 1/1 Points

Which special cases are used by the conflict-detection logic? 

Select all that apply.

- ☐ For an instruction with $I = 1$, its second operand is still treated as a register source for conflict detection.
- ☒ For a store instruction, the rd field is treated as a source register carrying the value to be stored.
- ☒ For a call instruction acting as the older producer, the destination is treated as ra.
- ☒ For a ret instruction, ra is treated as a source register.

✓ **Correct** 1/1 Points

Instruction A in OF is:

add r4, r2, 5 with $I = 1$.

Which older instruction B in EX/MA/RW creates a register conflict with A according to the conflict-detection algorithm? 

- ☒ sub r2, r8, r9
- ☐ st r2, 0[r8]
- ☐ cmp r2, r8
- ☐ add r5, r8, r9

✓ **Correct** 1/1 Points

Under the predict-not-taken branch-lock method, which behavior is correct when a branch reaches EX? 

- ☐ Hold the branch in EX until all younger instructions reach RW.
- ☐ Always fetch from the branch target first, and recover only if the branch is not taken.
- ☐ Always invalidate IF and OF, regardless of the branch outcome.
- ☒ If the branch is not taken, do nothing special; if it is taken, invalidate IF and OF and start fetching from the branch target.

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